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Optimal operational strategy of hybrid renewable energy system for rural electrification of a remote Algeria

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Abstract

Algeria has embarked on an ambitious renewable energy program in order to increase total food production. Energy planners for these remote locations have chosen diesel generation power technology because minimization of the initial capital cost of a power supply system is their top priority.

This paper proposes the use a PV-wind diesel generator hybrid system in order to determine the optimal configuration of renewable energy in Algeria and to compare the production cost of solar and wind power with its annual yield relevant to different regions in Algeria Djelfa, Ghardaïa, Adrar, tindouf and Tamanrasset on the best components and sizing with appropriate operating strategy to provide a cheap, efficient, reliable and cost-effective system. The Simulations were carried out using the HOMER program based on data obtained from the Algerian Meteorological Centre. For this approach, the energetic resources of sites where are implanted telecommunications systems and their consumption are supposed know.

Results show that, for Algeria, a PV diesel generator hybrid system is the most suitable solution in terms of economic performance and pollution.

However, the cost of production of solar and wind energy proved to be cheaper and more environmentally friendly than the energy produced from diesel generators.

key words: renewable energy, electric generator, radio telecommunications system, optimization, simulation.

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1. Introduction

Renewable Energy is a daily reality Development and the environment. In addition, much of the world will probably never connected to power grids whose extension is too expensive for isolated areas, sparsely populated or difficult access. And as we are interested in this Article possibilities installation systems telecommunications in these territories, the energy is therefore a renewable alternative to fossil fossils. An electric generator for supplying a telecommunications system using the renewable energy power must be optimized to exploit the maximum and efficient use in this application and subscriber expects such a system the same reliability that could offer an alternative source conventional energy.

The purpose of this work is to make available to users, an electric generator reliable, high efficiency and cost, using the energy renewable power systems radio. It is to find renewable energy sources in remote regions of Algeria, to supply our electric load, and optimize models appropriate to orders optimal.

2. Radio Telecommunications Systems

Telecommunication means, any type of transmission, emission or reception of signs, signals, writing, images, sounds or intelligence of any nature with or without support. For telecommunications facilities in rural areas, radio transmission is the most appropriate, it is based on the propagation of data without son.

3. Estimation of energy and choice of different sites in Algeria

The installation of telecommunication networks requires power continuously and without interruption. A study of the various renewable energy resources, in the case of this application, lead us to propose two inexhaustible renewable energy resources, abundant and profitable. These renewable resources are solar photovoltaic and wind energy. Our study focuses on the choice between these two resources, used individually or mixed, and is combined, in this case energy supply, and for different areas of Algeria.

choice of sites to study depends on the knowledge of the main characteristics of solar and wind fields. Algeria has a large pool solar almost all the national territory. Only the Sahara and the highlands have solar fields more important than the other regions according to a study done on two deposits in the country may be divided into four main areas:

- area of regions where the wind speed is between 5 m / s to 6 m / s,
- area where the regions have wind speeds between 4 m / s and 5 m / s,
- area regions have relatively low wind speeds, less than 4 m / s.

The sites chosen were: Adrar for the first zone, the second zone to Tindouf, Tamanrasset to the third zone and Ghardaïa is located in the fourth zone. And it is preferable to include a fifth site located in the highlands, Djelfa for example. The following table (Tab.1), data are presented on the wind speed and latitude of each site studied.

Table 1. values of wind speed and latitude for each site

	Wind speed	Latitude of site
Adrar	6 m/s à 7 m/s	27°53'
Tindouf	5 m/s à 6 m/s	27°40'
Tamanrasset	4 m/s à 5 m/s	22°47'
Djelfa	4 m/s	34°41'
Ghardaïa	3 m/s	32°23'

The solar field is given as a function of the latitude and varies with the variation of the latter. Thus it is given, on tab 2 and shown in Figure.1, the global irradiation in kWh / m² / day for five sites by decreasing latitude.

Table 2. global irradiation of each selected sites (kWh / m² / day)

Sites	Djelfa (34° 41')	Ghardaïa (32° 23')	Adrar (27° 53')	Tindouf (27° 40')	Tamanra sset (22° 47')
Mois					
january	1.712	1.840	2.086	2.097	2.354
february	2.404	2.532	2.771	2.782	3.025
march	3.557	3.667	3.864	3.873	4.059
april	5.015	5.080	5.183	5.187	5.263
may	5.752	5.758	5.746	5.745	5.696
june	5.844	5.819	5.751	5.747	5.640
july	5.996	5.985	5.941	5.938	5.855
august	4.999	5.038	5,093	5.095	5.120
september	4.234	4.334	4,510	4.518	4.677
October	2.915	3.046	3.290	3.302	3.544
November	2.071	2.213	2.486	2.498	2.781
December	1.619	1.753	2.013	2.025	2.298

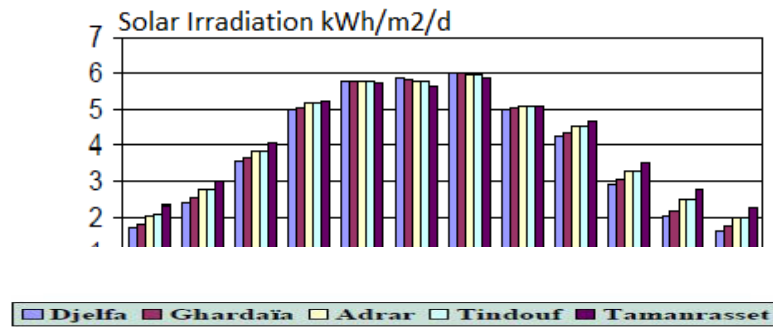


Fig. 1. Global solar irradiation for different sites

4. Modelling and Optimization of Generators of Renewable Energy

The electric generator using renewable energy resources and a radio communications system load, has a model shown schematically in Figure.2. It uses photovoltaic solar energy, wind power and storage.

Generator of renewable energy could of course be advantageously combined with other means of production, such as a heat generator Diesel, which would notably reduce the need for direct storage of electricity. It can be connected directly to the load, in the case where there is a cessation of production, as it can be connected in parallel with the other two generators of renewable energy, and this time it will be considered as an additional source.

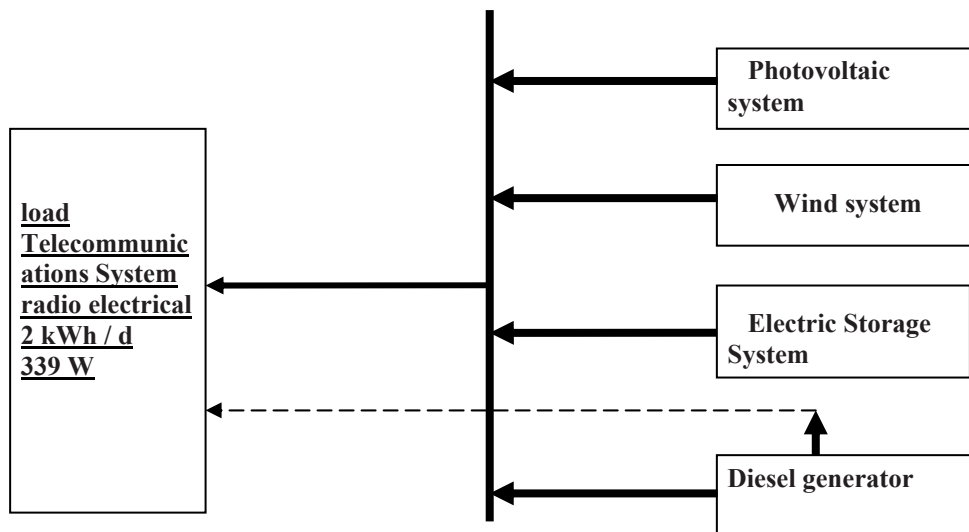


Fig. 2. Schematic model of the generator of renewable energy

4.1. Optimization of a generator of renewable energy supplying a radio telecommunication system

The optimization will not only get an absolute optimum, but also to find a good solution, and the guarantee of the absence of a solution significantly better. To achieve this goal after a reasonable computing time, it is necessary to use reliable methods.

Knowing the energy resources previously selected sites, as well as the energy consumption of radio telecommunications system, the software 'HOMER' optimizes various renewable generators

The load power is a sample of radio telecommunication systems, chosen for its energy consumption, the software 'HOMER' considered in its calculations.

Simulation using four electric energy production, for each site gives more optimized systems and each system uses a combination of energy resources.

The ability of both photovoltaic and wind renewable resources according to the wind speed curves shown in Figures 3 and 4.

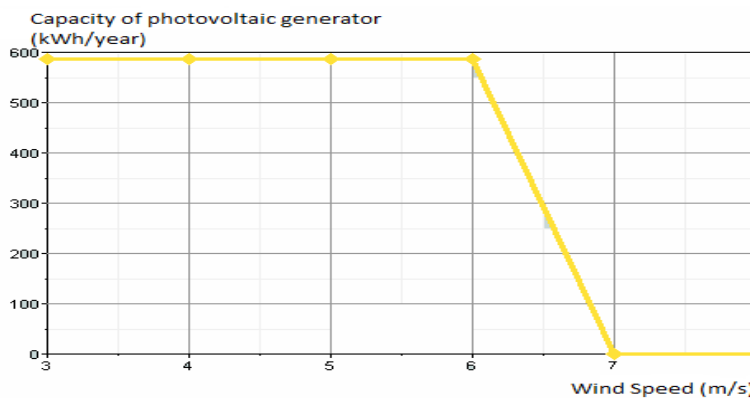


Fig. 3. Capacity of Photovoltaic Generator considered according wind speed

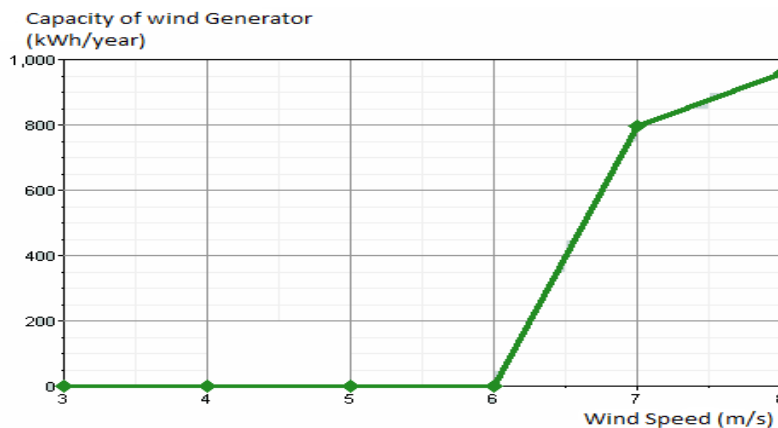


Fig. 4. Capacity of wind Generator considered according wind speed

4.1.1. Optimization of the generator in Ghardaïa site

The optimization of the electric generator, as designed and represented by the model chosen above gives the results presented and shown in Table 3.

Table 3. Optimization results of electric generator in ghardaïa site

	1 st optimized system	2 nd optimized system	3 rd optimized system	4 th optimized system
Pv	GhPv(1)	GhPv(2)	GhPv(3)	GhPv(4)
Eo		GhEo(2)		GhEo(4)
Ds	GhDs(1)	GhDs(2)		
St	GhSt(1)	GhSt(2)	GhSt(3)	GhSt(4)

Where Gh is the site of Ghardaia, followed by Pv: photovoltaic, Eo: Production wind Ds: Production by diesel or St: production storage system, and the number of optimized system.

The first generator uses a photovoltaic (GhPv (1)), a diesel (GHDs (1)) and output per electrical storage system (GHST (1)), and this is the most optimized system .

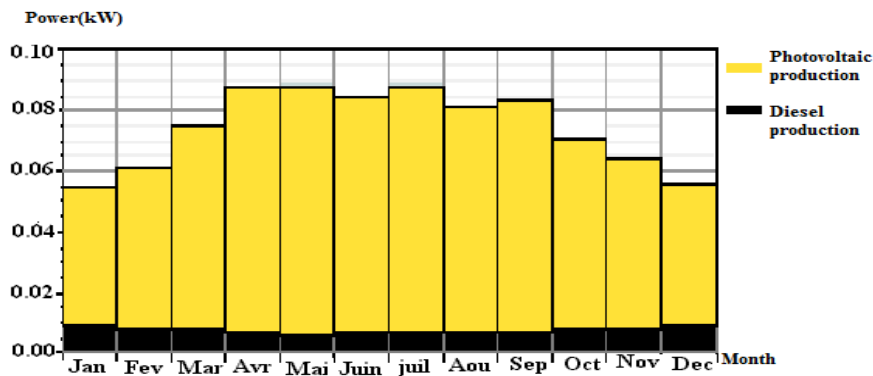


Fig. 5. power of the first optimized Electric generator in Ghardaïa site

It has 1% of power excess, 90% of the total production comes from PV generator and 10% remaining comes from the diesel generator.

According to the optimization study, the system of production of photovoltaic energy is always present, and it is present because the wind speed of the implantation site is low, it is preferable to use the solar energy resource.

4.1.2. Optimization of the generator in Djelfa site

The optimization of the electric generator in the Djelfa site, gives a system of photovoltaic (DjPv (1)), a diesel (DjDs (1)) and production by electrical storage system (DJST (1)), and this is the most optimized system. With the same results for Ghardaïa site because the wind speed is still low for this site ("4m / s).

4.1.3. Optimization of the generator in Tamanrasset site

The optimization of the electric generator in Tamanrasset site gives a system of photovoltaic (TAPV (1)), a diesel (TADS) (1) and production by electrical storage system (TAST (1)), and this is the most optimized system. With the same results for Ghardaïa site because the wind speed is still low for this site ("5m / s).

4.1.4. Optimization of the generator in Tindouf site

the optimization of the electric generator in Tindouf site gives a photovoltaic generation system and with the same results as the previous sites.

According to the optimization study, the energetic production system is supplying always by photovoltaic generator.

4.1.5. Optimization of the generator in Adrar site

The optimization of the electric generator in the site of Adrar, as designed and represented by the model chosen above gives the results presented and shown in Table 4. This site is the windiest, and the electric generator implanted in such a site will automatically have as a renewable resource of wind energy. We will confirm this hypothesis in the following results

The first electric generator is the best because it has 18% of energy excess and it is the lowest, 94% of the total production comes from wind turbine and the remaining 6% from the diesel generator. It presents as optimal renewable generation, wind power, and an excess of power higher than generators linked to other sites.

Table 4. Optimization results of electric generator in adrar site

	1 st optimized system	2 nd optimized system	3 rd optimized system	4 th optimized system	5 th optimized system	6 th optimized system
Pv		AdPv(2)	AdPv(3)	AdPv(4)		AdPv(6)
Eo	AdEo(1)		AdEo(3)	AdEo(4)	AdEo(5)	
Ds	AdDs(1)	AdDs(2)		AdDs(4)		
St	AdSt(1)	AdSt (2)	AdSt (3)	AdSt (4)	AdSt (5)	AdSt(6)

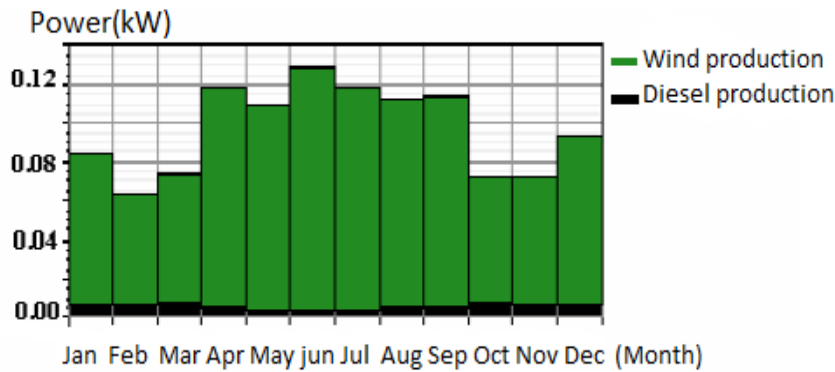


Fig 6. power of the first optimized Electric generator in Adrar site

4.2. Optimal system type

optimization results of the electrical generator using renewable energy with a diesel backup group show for all sites except for the last, that the more optimized system is the system using solar photovoltaic energy and the diesel production. These optimized systems have both: an excess production and lower system cost compared to other systems optimized.

The number of optimized systems varies from one site to another, depending on the potential of solar and wind field. It increases from a site to another less windy most windy.

The photovoltaic generator has an excess of low electric power.

The optimum system type for an electric generator using as renewable energy, solar photovoltaic and wind energy and associated diesel is generally represented for all sites in Figure 7.

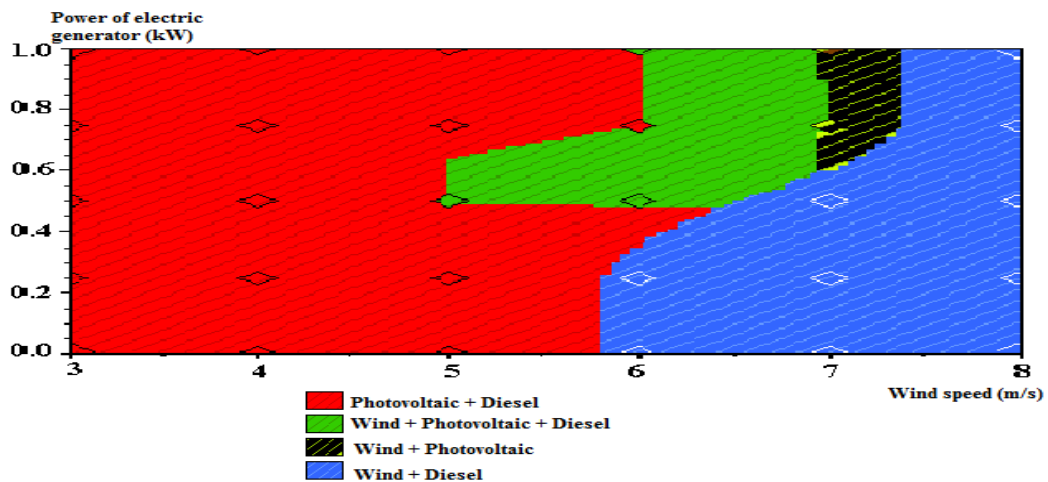


Fig.7. Optimal system type of electric generator using diesel as an additional source

The generator produces electrical energy from photovoltaic as the backup diesel generator for a wind speed of less than or equal to 6 m / s, and it was found from the figure that the previous representative photovoltaic production covers a large part electricity production mixed.

This generator produces electrical energy from wind power with the diesel as an adjunct for high wind speed, reaching 7 m / s. wind generation covers a smaller portion of the photovoltaic.

It has also been shown in the previous figure, that it is possible to associate the wind energy resource with the photovoltaic resource in sites where wind speed can reach 5 m / s and not more than 7m / s. and this mixed production as an adjunct diesel fuel is more important than the mixed production with renewable energies only (solar + wind).

5. multi source control circuit

The results of optimized generators using renewable energies obtained for each site, and providing an optimal type for the power equipment of telecommunications systems, to implement in Algeria and specifically in rural sites.

The control of the generator can be made according to the block diagram of Figure 8.

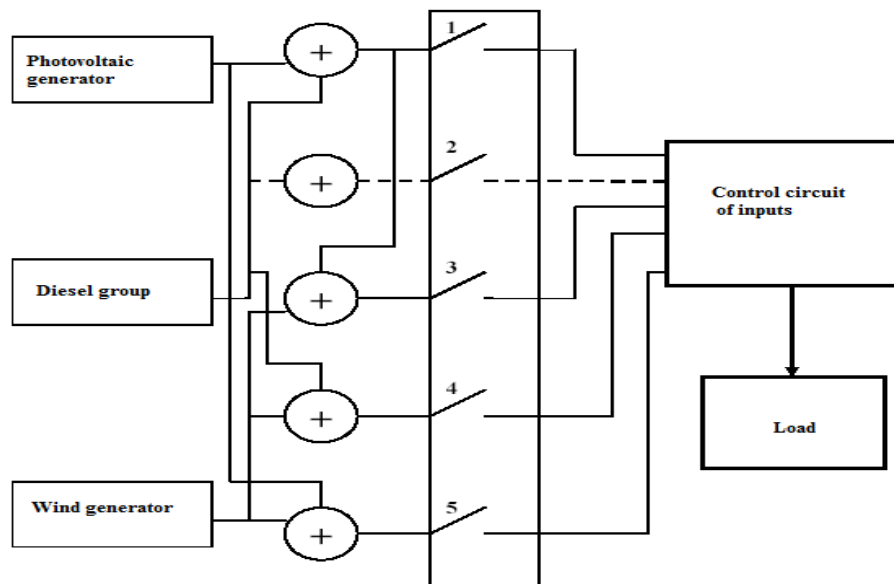


Fig.8. synoptic of electric generators using renewable energy

The control circuit can control inputs generators mixed. It opens a switch from five productions in order:1:

- 1: photovoltaic Production + diesel
- 2: diesel alone
- 3: photovoltaic Production + wind Production + diesel
- 4: wind Production + diesel
- 5: photovoltaic Production + wind Production

This command allows us to have all possible inputs, optimized generators, and have a constant supply of our load.

6. Conclusion

Renewable energy resources selected to supply a sample of telecommunication systems and the optimization of power generators using these resources helped us to have such an optimal system for supplying telecommunications equipment located in the middle rural Algeria.

These systems can be optimized subsequently controlled by a control circuit. So we can have depending on the availability of resources, one of the five combinations found in the optimal system type, and the telecommunications system will be powered permanently without any shortage and in all possible cases.

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